



Time Series Analysis

Chapter 4: Seasonality and forecasting: SARIMA, TBATS, Prophet



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Acronyms used in this material (1/2)

- ▣ **ACF**: Autocorrelation Function
- ▣ **AI**: Artificial Intelligence
- ▣ **AIC**: Akaike Information Criterion
- ▣ **AR**: AutoRegressive (model)
- ▣ **ARCH**: AutoRegressive Conditional Heteroskedasticity
- ▣ **ARIMA**: AutoRegressive Integrated Moving Average (model)
- ▣ **ARMA**: AutoRegressive Moving Average (model)
- ▣ **BIC**: Bayesian Information Criterion
- ▣ **CA**: Calendar Adjusted
- ▣ **DHR**: Dynamic Harmonic Regression
- ▣ **DM**: Diebold–Mariano (test)
- ▣ **ENTSO-E**: European Network of Transmission System Operators for Electricity
- ▣ **ETS**: Error, Trend, Seasonality (exponential smoothing state space models)
- ▣ **EU**: European Union
- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GDP**: Gross Domestic Product
- ▣ **HEGY**: Hylleberg–Engle–Granger–Yoo (seasonal unit-root test)
- ▣ **HICP**: Harmonised Index of Consumer Prices
- ▣ **HLN**: Harvey–Leybourne–Newbold (small-sample correction of the DM test)

Acronyms used in this material (2/2)

- ▣ **KPSS**: Kwiatkowski–Phillips–Schmidt–Shin (test)
- ▣ **LB**: Ljung–Box (test)
- ▣ **MA**: Moving Average
- ▣ **MAE**: Mean Absolute Error
- ▣ **MAPE**: Mean Absolute Percentage Error
- ▣ **MASE**: Mean Absolute Scaled Error
- ▣ **MSTL**: Multiple Seasonal-Trend decomposition using Loess
- ▣ **NSA**: Not Seasonally Adjusted
- ▣ **OCSB**: Osborn–Chui–Smith–Birchenhall (seasonal unit-root test)
- ▣ **PACF**: Partial AutoCorrelation Function
- ▣ **RMSE**: Root Mean Squared Error
- ▣ **SARIMA**: Seasonal ARIMA (model)
- ▣ **SCA**: Seasonally and Calendar Adjusted
- ▣ **SEATS**: Signal Extraction in ARIMA Time Series
- ▣ **STL**: Seasonal and Trend decomposition using Loess
- ▣ **TBATS**: Trigonometric seasonality, Box–Cox, ARMA errors, Trend, Seasonal components (model)
- ▣ **TRAMO-SEATS**: Time series Regression with ARIMA noise, Missing values and Outliers; Signal Extraction in ARIMA Time Series
- ▣ **US**: United States
- ▣ **UTC**: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ **VAT**: Value Added Tax

Today's question and route

- **Question:** many economic series repeat a pattern every year, week or day; how do we model this pattern, forecast with it, and decide which forecast is better?
 - ▶ three answers: **difference** the pattern away (SARIMA), **model** it with regressors (Fourier terms, TBATS, Prophet), or **remove** it (seasonal adjustment)
- **Route** of the chapter
 - ▶ seasonal differences and seasonal unit roots (HEGY, Canova–Hansen, OCSB)
 - ▶ $\text{SARIMA}(p, d, q)(P, D, Q)_s$: identification, estimation, diagnostics, forecasts; the airline model and Romanian GDP
 - ▶ seasonal adjustment in practice (X-13ARIMA-SEATS, Eurostat), Fourier terms and calendar effects (Orthodox Easter)
 - ▶ multiple seasonality in electricity load: MSTL, dynamic harmonic regression, TBATS, Prophet
 - ▶ forecast evaluation (time-series cross-validation, Diebold–Mariano test) and forecast combination
- It builds on [Chapter 0](#) (components, Holt–Winters, MASE) and [Chapters 2–3](#) (ARMA, unit roots, ARIMA): the same ideas, at the seasonal lag s
- Seminar 4 comes before this lecture: it gives the formulas; here we derive them and use them on data

Learning outcomes

- ▣ Recognise deterministic and stochastic seasonality and choose seasonal differencing with HEGY, Canova–Hansen and OCSB
- ▣ Write, identify, estimate, check and forecast a $\text{SARIMA}(p, d, q)(P, D, Q)_s$ model, and expand its polynomials
- ▣ Read seasonally adjusted and unadjusted official series correctly, and explain what X-13ARIMA-SEATS does
- ▣ Model calendar effects (working days, Orthodox Easter) and multiple seasonality with Fourier terms, MSTL, TBATS and Prophet
- ▣ Compare forecasts with time-series cross-validation, MASE and the Diebold–Mariano test, and combine them

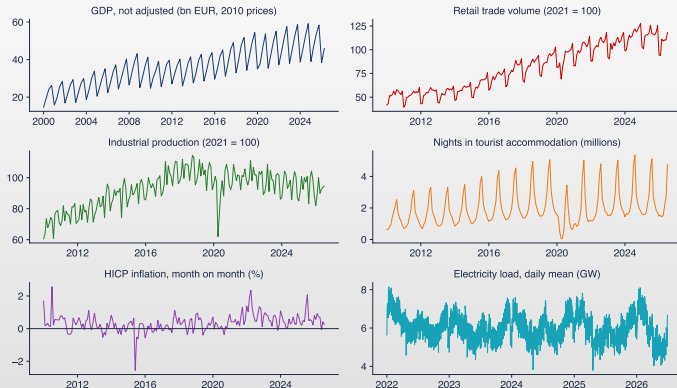
Reading and tools

- ▣ Textbook: [Huang and Petukhina \(2022\)](#), Ch. 4–5 (ARIMA and nonstationary models)
 - ▶ companion, free online: [Hyndman and Athanasopoulos \(2021\)](#), Ch. 9 (FPP3, Section 9.9), Ch. 10 (FPP3, Section 10.5), Ch. 12 (FPP3, Section 12.1) and Ch. 13
- ▣ The classic: [Box, Jenkins and Reinsel \(2008\)](#), Ch. 9 (seasonal models); seasonal adjustment: [Ladiray and Quenneville \(2001\)](#), [Dagum and Bianconcini \(2016\)](#), [Eurostat \(2024\)](#)
- ▣ Python Quantlets of this chapter: [Quantlets/Ch_04](#)
 - ▶ statsmodels (SARIMAX, MSTL, ExponentialSmoothing); optional packages tbats and prophet
- ▣ Lecture notebook: [open in Google Colab](#)
- ▣ Video course:  [Applied Time Series Analysis with Python](#)

Seasonality: definition and sources

- ▣ **Seasonality**: a pattern that repeats with a fixed, known **period** s (observations per cycle)
 - ▶ $s = 4$ (quarters), $s = 12$ (months), $s = 7$ (days in a week), $s = 24$ and $s = 168$ (hours in a day and in a week)
 - ▶ unlike a **cycle**, whose length is irregular and unknown ([Chapter 0](#))
- ▣ **Sources**
 - ▶ weather: heating in winter, the beach in August, harvests
 - ▶ the calendar: working days, public holidays, **moving holidays** such as Orthodox Easter
 - ▶ institutions and habits: the school year, tax deadlines, end-of-year bonuses, the working week
- ▣ **Additive** ($y_t = T_t + S_t + R_t$) or **multiplicative** ($y_t = T_t S_t R_t$, additive after logs): [Chapter 0](#); here we work with logs when the swings grow with the level

Seasonality in six Romanian series

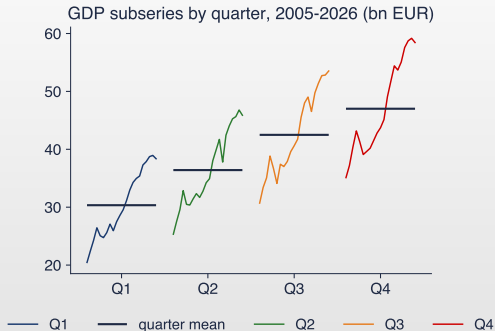
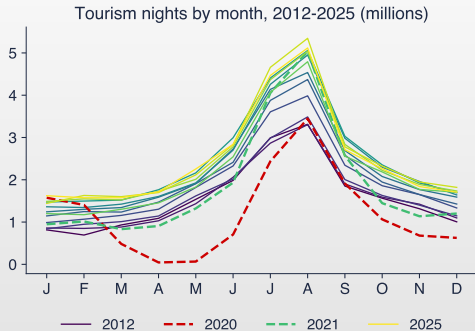


□ Eurostat: GDP (quarterly, not adjusted), retail trade, industrial production, tourism nights, HICP; ENTSO-E: daily electricity load

Interpreting the six series

- GDP: the mean of Q4 is 57% above the mean of Q1 (2000–2026); the swing grows with the level: multiplicative, take logs
 - ▶ quarter means (bn EUR, 2010 prices): 27.8, 33.6, 39.5, 43.6: construction and agriculture in Q2–Q3
- Tourism: August has 3.7 times the nights of January (2015–2019); the 2020 lockdown breaks the pattern
- HICP: monthly inflation is lower in June (-0.37% on average, 2010–2019, fresh food) and higher in October (0.49%)
- Electricity: two patterns at once, the week (dense oscillation) and the year (winter peaks): **multiple seasonality**, Section 8

Seasonal plot and subseries plot



□ Left: one line per year (seasonal plot); right: each quarter of GDP followed over the years, with its mean (subseries plot)

 TSA_ch4_seasonal_data

Interpreting the seasonal plots

- Tourism: one shape every year, a summer peak and a winter trough
 - ▶ 2019: 5.06 million nights in August, 1.47 million in January
 - ▶ April 2020: 0.045 million (1.76 million in April 2019): an outlier, not a new season
- GDP subseries: all four quarters rise in parallel
 - ▶ a stable seasonal shape on top of a common trend
 - ▶ if the lines crossed, the seasonal pattern would be changing: stochastic seasonality (next section)



Mamaia, on the Black Sea, September 2013

Photo: Razvan Socol (2013); CC BY-SA 4.0; Wikimedia

[Commons](#)

Recap: Seasonality in economic data

- Seasonality repeats with a known period s ; a cycle does not
- Weather, the calendar and institutions create it; moving holidays shift it between months
- Romanian GDP, tourism, inflation and electricity load are all seasonal; electricity has several periods at once
- Seasonal and subseries plots show whether the pattern is stable

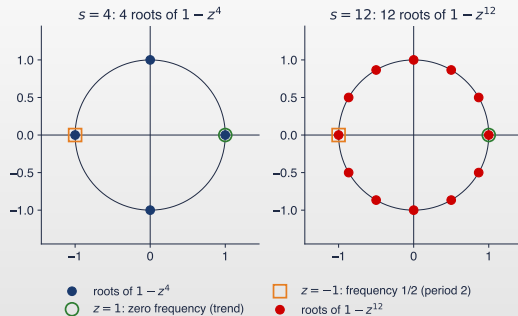
The seasonal difference

- **Seasonal difference:** $\Delta_s y_t = (1 - L^s)y_t = y_t - y_{t-s}$
 - ▶ with logs: $100 \Delta_4 \ln Y_t$ = growth over the same quarter of the previous year, in %
- A **fixed** seasonal pattern $S_t = S_{t-s}$ plus a linear trend: $y_t = a + bt + S_t + \varepsilon_t$
 - ▶ a : the intercept; b : the slope of the trend; S_t : the seasonal effect of period t
 - ▶ $\Delta_s y_t = bs + \varepsilon_t - \varepsilon_{t-s}$: Δ_s removes the pattern and turns the trend into a constant
 - ▶ but it creates the non-invertible MA term $\varepsilon_t - \varepsilon_{t-s}$: **seasonal over-differencing** (as in [Chapter 3](#))
- $\Delta\Delta_s = (1 - L)(1 - L^s) = 1 - L - L^s + L^{s+1}$: needed when the trend is stochastic as well
 - ▶ the order of the two differences does not matter: the operators commute

Deterministic and stochastic seasonality

- **Deterministic:** $y_t = \sum_{j=1}^s \gamma_j D_{jt} + u_t$, with $D_{jt} = 1$ in season j and u_t stationary
 - ▶ the seasonal means γ_j never change; the right tool: seasonal dummies or Fourier terms (Section 7)
- **Stochastic:** the **seasonal random walk** $y_t = y_{t-s} + \varepsilon_t$
 - ▶ each season is its own random walk: s separate walks interleaved
 - ▶ $\text{Var}(y_t)$ grows without bound; the pattern drifts, and “summer can become winter”
 - ▶ the right tool: the seasonal difference Δ_s
- The choice matters: differencing a deterministic pattern over-differences; modelling a stochastic one with dummies leaves a unit root in the residuals

The roots of $1 - z^s$: seasonal unit roots

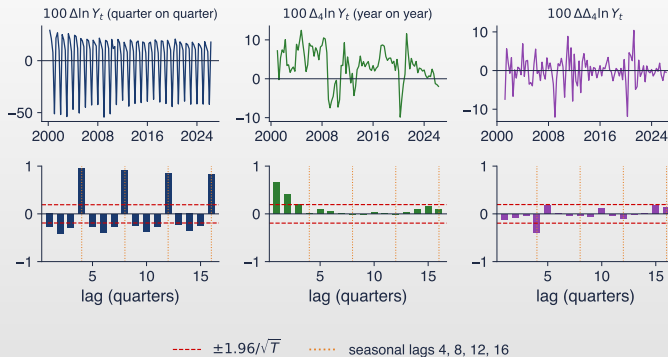


□ $1 - z^4 = (1 - z)(1 + z)(1 + z^2)$: roots 1, -1 , $\pm i$, all on the unit circle

Interpreting the seasonal roots

- Δ_s imposes s unit roots at once, one for each **frequency** k/s , $k = 0, \dots, s - 1$
 - ▶ quarterly: $z = 1$ (frequency 0, the trend), $z = -1$ (half a cycle per quarter: period 2 quarters), $z = \pm i$ (period 4 quarters: the annual cycle)
- A series may have some of these roots and not others: then Δ_4 over-differences at the frequencies it does not need
- Seasonal unit-root tests ask, frequency by frequency, which roots are present: the HEGY test (next slides)

Romanian GDP: regular and seasonal differences



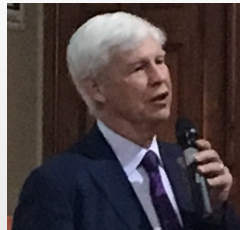
- In of real GDP, not adjusted, 106 quarters to 2026Q2; top: three transformations; bottom: their sample ACF up to 16 quarters

Interpreting the three transformations

- $\Delta \ln Y$: dominated by the season: ACF 0.95 at lag 4 and 0.90 at lag 8, decaying very slowly; sd 26.5%
 - ▶ the seasonal analogue of the slow ACF decay of a random walk ([Chapter 3](#))
- $\Delta_4 \ln Y$ (year-on-year growth): sd 4.4%; ACF 0.66 at lag 1, 0.01 at lag 4: an AR-type pattern
- $\Delta\Delta_4 \ln Y$: ACF -0.11 at lag 1 and -0.40 at lag 4 (band ± 0.20)
 - ▶ one negative spike at the seasonal lag: a seasonal MA(1), the signature of the airline model (Section 3)

The HEGY test

- **Hylleberg, Engle, Granger and Yoo (1990)**: one regression tests every seasonal root of a quarterly series
 - ▶ filters: $y_{1t} = (1 + L + L^2 + L^3)y_t$, $y_{2t} = -(1 - L + L^2 - L^3)y_t$,
 $y_{3t} = -(1 - L^2)y_t$
- $\Delta_4 y_t = \pi_1 y_{1,t-1} + \pi_2 y_{2,t-1} + \pi_3 y_{3,t-2} + \pi_4 y_{3,t-1} + \text{deterministic terms} + \text{lags of } \Delta_4 y_t + \varepsilon_t$
 - ▶ y_{1t}, y_{2t}, y_{3t} : the series filtered so that each keeps one seasonal frequency;
 π_k : the coefficient of each frequency
 - ▶ H_0 : $\pi_1 = 0$ (root 1), t -test; $\pi_2 = 0$ (root -1), t -test; $\pi_3 = \pi_4 = 0$ (roots $\pm i$), F -test
 - ▶ non-standard distributions, as for Dickey–Fuller: critical values from tables or by simulation
- Rejecting all three: no seasonal differencing needed; rejecting none: Δ_4 is justified



Robert Engle, a co-author of HEGY, 2017

Photo: Econterms (2017); CC BY-SA 4.0;
 Wikimedia Commons

Canova–Hansen and OCSB

- ▣ **Canova–Hansen** (Canova and Hansen (1995)): the null is **stable** (deterministic) seasonality
 - ▶ the seasonal analogue of KPSS (Chapter 3): rejecting means the pattern changes over time
- ▣ **OCSB** (Osborn, Chui, Smith and Birchenhall (1988)): the null is a seasonal unit root; a regression of $\Delta\Delta_s y_t$ on $\Delta_s y_{t-1}$ and Δy_{t-s}
 - ▶ the default seasonal test of `auto_arima` in the Python package `pmdarima` (`nsdiffs`)
- ▣ As in Chapter 3: tests with opposite nulls answer different questions; use them together
 - ▶ and look at the data: seasonal plots, the ACF at lags $s, 2s, 3s$, and the size of $\hat{\Theta}$ after Δ_s ($\hat{\Theta} \approx -1$ signals over-differencing)

Seasonal unit-root tests on Romanian data

HEGY, ln GDP, 2000–2019	statistic	5% critical value	decision
$t(\pi_1)$, root 1	−1.97	−3.39	not rejected
$t(\pi_2)$, root −1	−2.19	−2.77	not rejected
$F(\pi_3, \pi_4)$, roots $\pm i$	6.12	6.64	not rejected

Series (logs, 2005–2019)	D by OCSB	D by Canova–Hansen	d by KPSS
GDP (quarterly)	1	0	1
Retail trade	0	0	1
Industrial production	0	0	1
Tourism nights	0	0	1
HICP	0	0	2

- HEGY: constant, seasonal dummies and trend, 1 lag of $\Delta_4 y$ (AIC), $n = 75$; critical values from 4 000 simulations of $\Delta_4 y_t = \varepsilon_t$
- Interpretation: for GDP, HEGY and OCSB support Δ_4 while Canova–Hansen does not reject a stable pattern; with 20 years of quarters both views fit, so we use Δ_4 and let $\hat{\Theta}$ say how stable the pattern is (Section 5)
- Monthly series: both tests choose $D = 0$, a stable pattern; HICP needs $d = 2$ in logs: inflation itself is close to a unit root ([Chapter 3](#))

Recap: Seasonal differencing and seasonal unit roots

- ▣ $\Delta_s = 1 - L^s$ has s unit roots on the circle, one per seasonal frequency
- ▣ Deterministic seasonality: dummies or Fourier terms; stochastic seasonality: Δ_s
- ▣ HEGY and OCSB test for seasonal unit roots; Canova–Hansen tests for a stable pattern
- ▣ Romanian GDP: $\Delta\Delta_4 \ln Y$ leaves one spike at lag 4, a seasonal MA

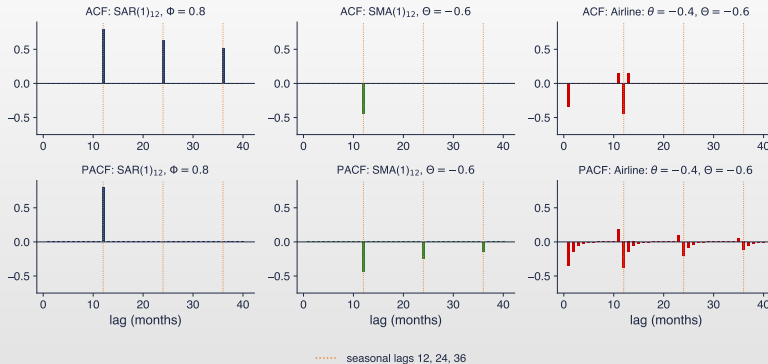
The SARIMA(p, d, q)(P, D, Q) $_s$ model

- $\phi(L) \Phi(L^s) (1 - L)^d (1 - L^s)^D y_t = c + \theta(L) \Theta(L^s) \varepsilon_t, \varepsilon_t \sim \text{WN}(0, \sigma^2)$
 - ▶ regular polynomials $\phi(L)$ (order p), $\theta(L)$ (order q), as in [Chapter 2](#)
 - ▶ seasonal polynomials in L^s (orders P, Q): $\Phi(L^s) = 1 - \Phi_1 L^s - \dots - \Phi_P L^{Ps}$, $\Theta(L^s) = 1 + \Theta_1 L^s + \dots + \Theta_Q L^{Qs}$
 - ▶ d, D : numbers of regular and seasonal differences, usually 0 or 1
- Why multiply? [Box, Jenkins and Reinsel \(2008\)](#): in a monthly series, January depends on December (the regular part) and on last January (the seasonal part)
 - ▶ a two-way table: months in the columns, years in the rows; one model along each direction
- Stationarity: the roots of $\phi(z)\Phi(z^s)$ outside the unit circle; invertibility: the roots of $\theta(z)\Theta(z^s)$ outside; for a seasonal AR(1) this means $|\Phi| < 1$

Worked example: the airline model

- SARIMA(0, 1, 1)(0, 1, 1)₁₂ with $\theta = -0.4$, $\Theta = -0.6$: $(1 - L)(1 - L^{12})y_t = (1 - 0.4L)(1 - 0.6L^{12})\varepsilon_t$
 - ▶ left: $y_t - y_{t-1} - y_{t-12} + y_{t-13}$; right: $\varepsilon_t - 0.4\varepsilon_{t-1} - 0.6\varepsilon_{t-12} + 0.24\varepsilon_{t-13}$
 - ▶ the coefficient of ε_{t-13} , $\theta\Theta = 0.24$, is not a new parameter: two parameters plus σ^2
- $w_t = \Delta\Delta_{12}y_t$ is an MA(13) with only four nonzero weights; its autocovariances ([Chapter 2](#)):
 - ▶ $\gamma(0) = \sigma^2(1 + \theta^2)(1 + \Theta^2) = 1.5776\sigma^2$
 - ▶ $\rho_1 = \theta/(1 + \theta^2) = -0.345$; $\rho_{12} = \Theta/(1 + \Theta^2) = -0.441$; $\rho_{11} = \rho_{13} = \rho_1\rho_{12} = 0.152$
 - ▶ all other autocorrelations are zero

Theoretical ACF and PACF of three seasonal models



□ Monthly, $s = 12$; the airline panel is the stationary part $w_t = \Delta\Delta_{12}y_t$

Interpreting the seasonal patterns

- Seasonal AR(1): the ACF decays geometrically at lags 12, 24, 36 (0.8, 0.64, 0.512); the PACF has one spike at lag 12
 - ▶ the AR rule of [Chapter 2](#), read only at multiples of s
- Seasonal MA(1): the ACF cuts off after lag 12; the PACF decays at 12, 24, 36 (-0.242 at lag 24)
- Airline: spikes at 1 (-0.345) and 12 (-0.441), with “satellites” at 11 and 13 (0.152)
 - ▶ the satellites are the product $\rho_1\rho_{12}$: the fingerprint of the multiplicative structure

Identification of SARIMA models

After differencing	ACF	PACF
seasonal AR(P)	decays at $s, 2s, \dots$	cuts off after lag Ps
seasonal MA(Q)	cuts off after lag Qs	decays at $s, 2s, \dots$
regular part	lags $1, 2, \dots$ as in Chapter 2	lags $1, 2, \dots$ as in Chapter 2
product of both	satellites at $ks \pm j$	satellites at $ks \pm j$

□ Steps

- ▶ 1. Stabilise the variance (logs or Box–Cox, Box and Cox (1964)); 2. choose D (seasonal plot, HEGY, OCSB), then d (Chapter 3)
- ▶ 3. Read P, Q at the seasonal lags and p, q at the first lags; 4. keep the orders small: $P, Q \leq 1$ almost always suffice

Estimation and model choice

- **Maximum likelihood** of the differenced series (state-space form, `statsmodels SARIMAX`)
 - ▶ $s + 1$ observations are lost to $\Delta\Delta_s$; the constant is dropped when $d + D \geq 2$ (it would mean a quadratic trend)
- **Information criteria:** $AICc = AIC + \frac{2k(k+1)}{n-k-1}$ and BIC, compared **only** across models with the same d and D
 - ▶ k : the number of estimated parameters; n : the number of observations after differencing
 - ▶ differencing changes the data on which the likelihood is computed
- **Automatic SARIMA:** a stepwise search over (p, q, P, Q) by AICc after choosing d and D by tests ([Hyndman and Khandakar \(2008\)](#))
 - ▶ in Python: `pmdarima.auto_arima(y, m=12)`; always check the residuals and compare with a benchmark
- **Diagnostics:** Ljung–Box with lags beyond s (e.g. $m = 2s$) and $m - p - q - P - Q$ degrees of freedom ([Ljung and Box \(1978\)](#))

Forecasting with SARIMA

- As for ARIMA (Chapter 3): write the model as an equation for y_t ; replace future shocks by 0, future values by their forecasts, past shocks by residuals
 - ▶ airline: $\hat{y}_{T+h} = \hat{y}_{T+h-1} + \hat{y}_{T+h-s} - \hat{y}_{T+h-s-1} + (\text{MA terms while } h \leq s + 1)$
- **Eventual forecast function** of the airline model: a fixed seasonal pattern on a straight line
 - ▶ the pattern is an exponentially weighted average of past patterns, with weight $1 + \Theta$ on the most recent year
 - ▶ $\Theta \rightarrow -1$: a fixed (deterministic) pattern; $\Theta = 0$: last year's pattern repeated (seasonal naive)
- Intervals widen with h (two unit roots); for $\ln y$, $\exp(\hat{y}_{T+h})$ is the forecast **median**
 - ▶ the mean is $\exp(\hat{y}_{T+h} + \sigma_h^2/2)$, slightly higher; σ_h^2 : the forecast error variance at horizon h

Worked example: forecasting a seasonal MA

- Quarterly SARIMA(0,0,0)(0,1,1)₄: $y_t = y_{t-4} + \varepsilon_t + \Theta \varepsilon_{t-4}$, $\Theta = -0.6$
 - ▶ last year: $y_{T-3}, \dots, y_T = 100, 120, 130, 140$; residuals $\hat{\varepsilon}_{T-3}, \dots, \hat{\varepsilon}_T = 2, -1, 0, 3$
- $\hat{y}_{T+1} = y_{T-3} + \Theta \hat{\varepsilon}_{T-3} = 100 - 0.6 \cdot 2 = 98.8$
 - ▶ $\hat{y}_{T+2} = 120.6$, $\hat{y}_{T+3} = 130.0$, $\hat{y}_{T+4} = 138.2$
- $\hat{y}_{T+5} = \hat{y}_{T+1} = 98.8$: after one year the MA term is gone and the forecast pattern repeats
 - ▶ a large positive surprise last season ($\hat{\varepsilon}_T = 3$) lowers the forecast for that season by $0.6 \cdot 3$: partial correction

Recap: SARIMA models

- ▣ SARIMA multiplies regular and seasonal polynomials: $\phi(L)\Phi(L^s)\Delta^d\Delta_s^D y_t = \theta(L)\Theta(L^s)\varepsilon_t$
- ▣ Identification: the [Chapter 2](#) rules at lags $s, 2s, \dots$, plus satellites at $ks \pm j$
- ▣ Compare AICc only for the same d and D ; check Ljung–Box beyond lag s
- ▣ The airline forecast: a seasonal pattern on a line, updated with weight $1 + \Theta$

Box and Jenkins and the airline passengers

- **Box, Jenkins and Reinsel (2008)** (first edition 1970) modelled the monthly totals of international airline passengers, 1949–1960 (144 months)
 - ▶ growth plus a summer peak whose size grows with the level
 - ▶ their model, $SARIMA(0, 1, 1)(0, 1, 1)_{12}$ for $\ln y_t$, has carried the name “airline model” ever since
- Why it lasted
 - ▶ two parameters fit many seasonal series: trade, production, retail sales, energy
 - ▶ it is the reference model of the regARIMA step of seasonal-adjustment software (Section 6)

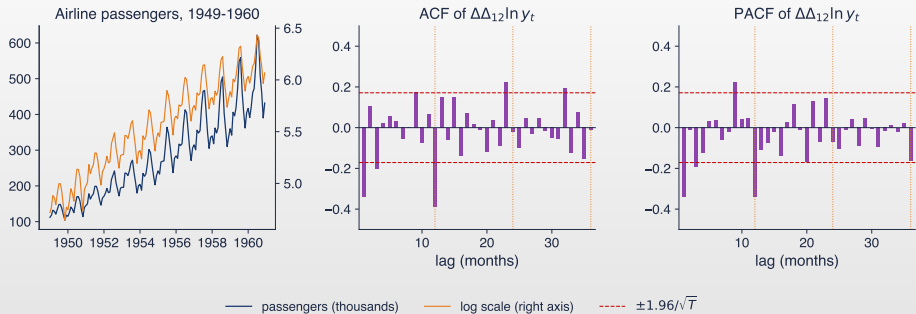


The first Boeing 707s for Pan American, Seattle, 1958

Photo: unknown author (1958); public domain; [Wikimedia](#)

[Commons](#)

Identification of the airline model

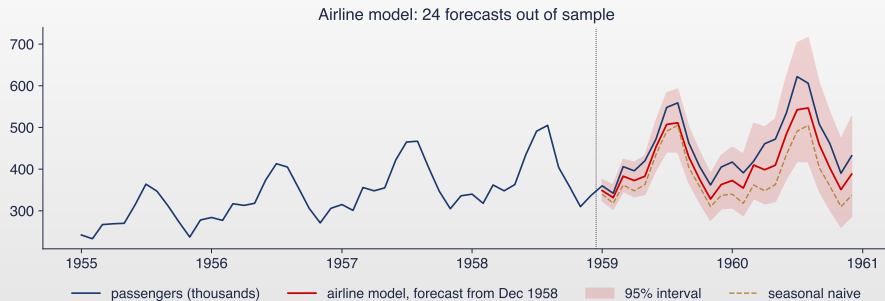


□ Left: passengers and their log; middle and right: ACF and PACF of $w_t = \Delta\Delta_{12}\ln y_t$, 131 observations

Interpreting the airline correlograms

- The log makes the summer swing constant: the multiplicative pattern becomes additive
- ACF of w_t : -0.34 at lag 1 and -0.39 at lag 12, outside the band ± 0.17
 - ▶ a regular MA(1) and a seasonal MA(1); lag 3 (-0.20) is borderline
- The PACF decays at lags 1, 2, 3 and 12, 24: consistent with MA terms, not AR terms
- Conclusion: SARIMA(0, 1, 1)(0, 1, 1)₁₂, exactly the model of Box and Jenkins

Airline model: forecasts out of sample



Estimated on 1949–1958, forecasts for the 24 months of 1959–1960, back-transformed from $\ln y$

Interpreting the airline forecasts

- 1949–1958: $\hat{\theta} = -0.342$ (0.087), $\hat{\Theta} = -0.541$ (0.105); full sample: $\hat{\theta} = -0.402$ (0.073), $\hat{\Theta} = -0.557$ (0.096)
 - ▶ residual sd 3.7% per month; Ljung–Box $Q^*(24)$ on 22 degrees of freedom, $p = 0.37$
- MAPE over 1959–1960: 8.5% for the airline model, 15.5% for the seasonal naive forecast (which ignores growth)
- December 1960: forecast 388, interval [286, 526], actual 432: inside, but the model under-predicts the 1959–1960 growth
 - ▶ the interval width grows from 51 (first month) to 239 (24th month): two unit roots

SARIMA models for Romanian GDP

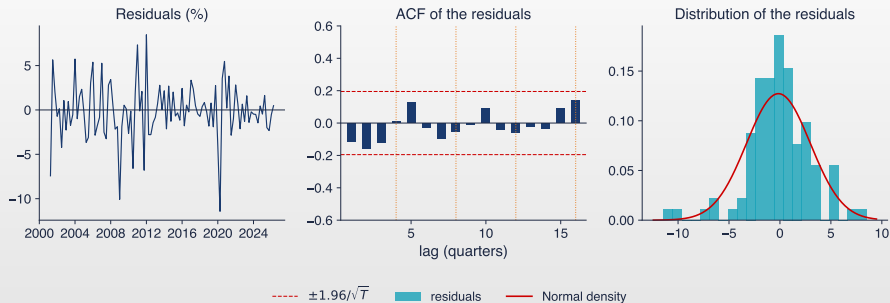
Model	AICc	BIC	LB(8), p	$\hat{\sigma}$ (%)
(0, 1, 1)(0, 1, 1) ₄	520.1	527.6	0.157	3.05
(0, 1, 0)(0, 1, 1) ₄	519.7	524.8	0.286	3.08
(1, 1, 0)(0, 1, 1) ₄	520.7	528.3	0.156	3.06
(1, 1, 1)(0, 1, 1) ₄	518.3	528.4	0.467	2.98
(0, 1, 2)(0, 1, 1) ₄	518.8	528.8	0.575	3.00
(0, 1, 1)(1, 1, 0) ₄	529.9	537.5	0.003	3.22
(0, 1, 1)(1, 1, 1) ₄	522.2	532.2	0.118	3.05
(0, 1, 1)(0, 1, 0) ₄	545.1	550.2	0.005	3.52

- $y_t = 100 \ln Y_t$, real GDP not adjusted, 2000Q1–2026Q2, $T = 106$; maximum likelihood; all with $d = D = 1$, so the criteria are comparable
- Ljung–Box on 8 lags with $8 - p - q - P - Q$ degrees of freedom

Interpreting the model table

- BIC chooses $(0, 1, 0)(0, 1, 1)_4$: only the seasonal MA term; AICc chooses $(1, 1, 1)(0, 1, 1)_4$
 - ▶ the AICc model has $\hat{\phi} = 0.78$ and $\hat{\theta} = -0.94$: almost a common factor ([Chapter 2](#)), little gain
 - ▶ the airline model: $\hat{\theta} = -0.171$ (0.109), not significant; $\hat{\Theta} = -0.585$
- A seasonal AR instead of the seasonal MA, or no seasonal term at all: much worse criteria and correlated residuals
- We keep the BIC model: one parameter, $\hat{\Theta} = -0.577$ (0.075), $\hat{\sigma} = 3.08\%$

Diagnostics of the GDP model

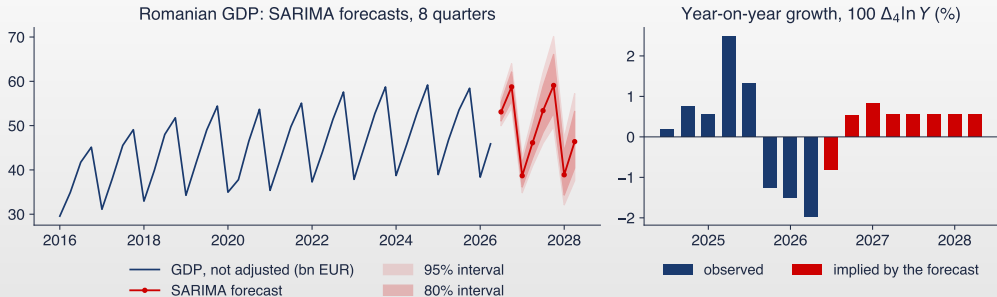


□ SARIMA(0, 1, 0)(0, 1, 1)₄ for $100 \ln Y_t$: residuals, their ACF and their distribution

Interpreting the GDP diagnostics

- No autocorrelation left: $Q^*(8) = 8.55$ on 7 degrees of freedom, $p = 0.286$; $Q^*(12)$: $p = 0.526$
 - ▶ no spike at the seasonal lags 4, 8, 12, 16: the seasonal structure is captured
- Not Normal: Jarque–Bera 22.0 ($p < 0.001$); the largest residual is -11.4% in 2020Q2 (-3.6 standard deviations): the lockdown
 - ▶ without 2020, Jarque–Bera falls to 13.4: an additive-outlier dummy would be the next step

Forecasts of Romanian GDP



□ SARIMA(0, 1, 0)(0, 1, 1)₄, 8 quarters from 2026Q3 to 2028Q2, 80% and 95% intervals; right: implied year-on-year growth

Interpreting the GDP forecasts

- Last value: 45.9 bn EUR in 2026Q2; forecasts 53.1, 58.7, 38.7, 46.1 bn EUR for the next four quarters
 - ▶ 95% interval for the first quarter [50.0, 56.4], for the eighth [37.6, 57.2]
- Implied year-on-year growth: -0.81% , 0.53% , 0.83% , 0.55% , then a constant 0.55%
 - ▶ the model extrapolates the recent weak growth; it knows nothing about fiscal policy or EU funds
- The seasonal pattern of the forecast is a weighted average of past years, with weight $1 + \hat{\Theta} = 0.42$ on the newest year

Recap: Airline passengers and Romanian GDP

- ▣ Logs, then $\Delta\Delta_s$, then MA terms at lags 1 and s : the airline recipe works for both series
- ▣ For GDP, BIC drops the regular MA term; AICc picks a near-redundant ARMA(1,1)
- ▣ Residuals are clean except for the 2020 outlier
- ▣ Forecasts repeat a smoothed seasonal pattern on a trend line, with widening intervals

Why official statistics adjust for seasonality

- ▣ Users want to know whether the economy **grew this quarter**, not whether Q3 is bigger than Q2
 - ▶ 2026Q1: unadjusted GDP changed by -42.1% (log points) from the previous quarter; the adjusted series by -0.06%
 - ▶ the first number is pure season; only the second says something about the business cycle
- ▣ **Seasonal adjustment**: estimate the seasonal and calendar components and remove them: $y_t^{SA} = y_t - \hat{S}_t - \hat{C}_t$ (or $y_t / (\hat{S}_t \hat{C}_t)$)
 - ▶ the classical decomposition and STL of [Chapter 0](#) are simple versions of this
- ▣ For forecasting with SARIMA we use the **unadjusted** series: adjustment filters change the autocorrelations and are revised when new data arrive

X-11 and X-13ARIMA-SEATS

- **X-11** (US Census Bureau, 1960s; [Ladiray and Quenneville \(2001\)](#)): iterated moving averages
 - ▶ a centred 2×12 moving average for the trend, then 3×3 and 3×5 moving averages of each month across years for the seasonal factors
- **X-12-ARIMA** ([Findley et al. \(1998\)](#)) and **X-13ARIMA-SEATS** add a **regARIMA** pre-adjustment:
 - ▶ regression with SARIMA errors (often the airline model) for outliers, working days, Easter
 - ▶ ARIMA forecasts extend the series, so the symmetric filters also work at the end: smaller revisions
 - ▶ then X-11 filters or SEATS (model-based decomposition of the ARIMA model)
- EU: Eurostat and the national institutes follow [Eurostat \(2024\)](#), with the JDemetra+ software (X-13 and TRAMO-SEATS)



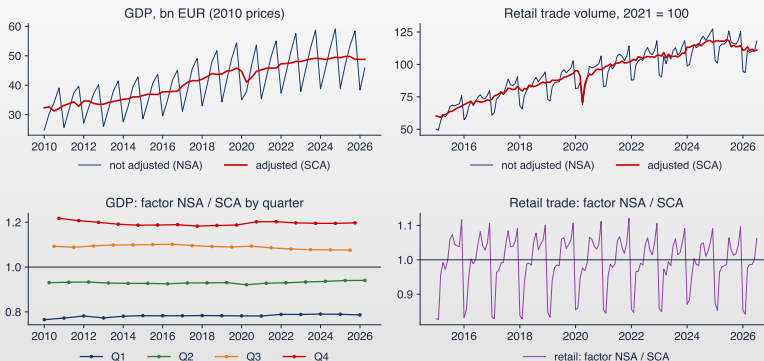
US Census Bureau, Suitland, Maryland:
the home of X-11 and X-13

Photo: [United States Census Bureau \(2007\)](#);
public domain; [Wikimedia Commons](#)

Eurostat series: NSA, CA, SA, SCA

- Each series comes in several versions (dimension `s_adj`):
 - ▶ **NSA**: not seasonally adjusted (the raw data)
 - ▶ **CA**: calendar adjusted only (working days, holidays)
 - ▶ **SA**: seasonally adjusted; **SCA**: seasonally and calendar adjusted
- Which one for which question
 - ▶ quarter-on-quarter or month-on-month growth: SCA
 - ▶ year-on-year growth: NSA or CA (the seasonal pattern cancels, the calendar does not)
 - ▶ SARIMA modelling and forecasting: NSA, with calendar regressors
- The annual totals of NSA and SCA are close (2024: 196.4 and 196.4 bn EUR): adjustment moves output between quarters, it does not create it

Unadjusted and adjusted series



- Eurostat NSA and SCA: GDP (top left) and retail trade (top right); bottom: the implied seasonal and calendar factors NSA/SCA

Interpreting the adjusted series

- GDP factors 2015–2019: Q1 0.78, Q2 0.93, Q3 1.10, Q4 1.19: Q1 output is about a fifth below an average quarter
 - ▶ the factors move slowly over the years: the stochastic seasonality that $\hat{\Theta}$ measures
- Retail trade: December factor 1.10, January 0.84 (2016–2019): Christmas shopping, then the January low
- The retail factor also jumps from month to month because of working days and Easter: the calendar part of SCA (Section 7)
- The adjusted GDP shows the 2020 fall and the 2025–2026 stagnation that the raw series hides

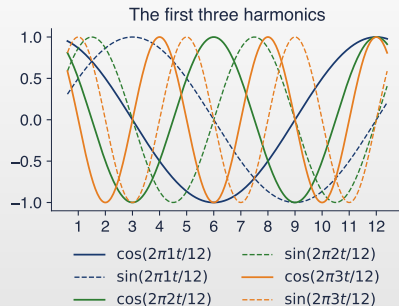
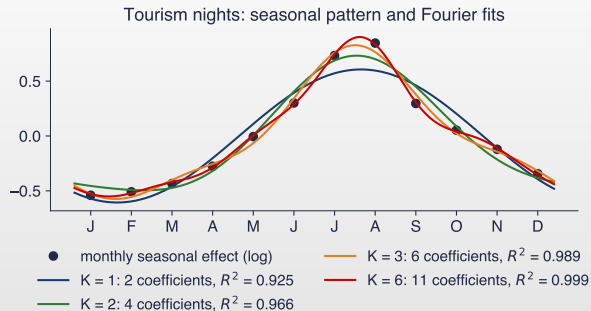
Recap: Seasonal adjustment

- Seasonal adjustment removes $\hat{S}_t$ and $\hat{C}_t$ to show the trend and the cycle
- X-13ARIMA-SEATS: regARIMA pre-adjustment (outliers, calendar, forecasts), then X-11 or SEATS; in the EU, JDemetra+
- Quarter-on-quarter growth: SCA; year-on-year growth: NSA or CA; SARIMA forecasting: NSA

Seasonal dummies and Fourier terms

- **Dummies:** $y_t = \beta_0 + \sum_{j=2}^s \gamma_j D_{jt} + u_t$: $s - 1$ coefficients, any shape
 - ▶ β_0 : the level of season 1; γ_j : the difference between season j and season 1
 - ▶ too many for $s = 52$ weeks, $s = 168$ hours or $s = 365$ days; impossible for $s = 365.25$
- **Fourier terms** for a period m : $\sum_{k=1}^K \left[\alpha_k \sin \frac{2\pi kt}{m} + \beta_k \cos \frac{2\pi kt}{m} \right]$
 - ▶ α_k, β_k : the coefficients of the k -th harmonic, a wave with k cycles per period; K : the number of sine–cosine pairs
 - ▶ $2K$ coefficients; $K = m/2$ reproduces the dummies exactly (for even m); a small K gives a smooth pattern
 - ▶ m may be non-integer, and several periods can be combined: e.g. $K = 5$ for $m = 24$ and $K = 4$ for $m = 168$ give 18 regressors instead of $23 + 167$
- K is chosen by AICc or by cross-validation; the same terms are inside TBATS and Prophet (Section 9)

Fourier approximation of a seasonal pattern



□ Monthly seasonal effect of $\ln(\text{tourism nights})$, 2012–2019, after removing a 13-month centred mean; fits with $K = 1, 2, 3, 6$

 TSA_ch4_calendar_fourier

Interpreting the Fourier fits

- $K = 1$ (two coefficients) already gives $R^2 = 0.925$: the summer peak is close to a single wave
 - ▶ $K = 2$: 0.966; $K = 3$: 0.989 with six coefficients instead of eleven
- $K = 6$ (11 coefficients) interpolates the 12 monthly points: the same as 11 dummies, $R^2 = 0.999$
- Higher harmonics add sharper detail: the August peak needs $K \geq 3$

Calendar effects and Orthodox Easter

- **Working days:** a month with more Monday–Friday non-holiday days produces and sells more
 - ▶ known in advance: a regressor that can be used in forecasts
- **Moving holidays:** Orthodox Easter falls between 4 April and 8 May
 - ▶ 5 May 2024, 20 April 2025, 12 April 2026, 2 May 2027: shopping shifts between March, April and May
 - ▶ in Python: `dateutil.easter.easter(y, EASTER_ORTHODOX)`; Pentecost is 49 days later
- **Easter regressor** (X-13 `easter[w]`): E_t = share of the w days before Easter Sunday that fall in month t
 - ▶ $w = 10$: Easter on 20 April gives $E_{\text{April}} = 1$; on 5 May, $E_{\text{April}} = 0.6$ and $E_{\text{May}} = 0.4$



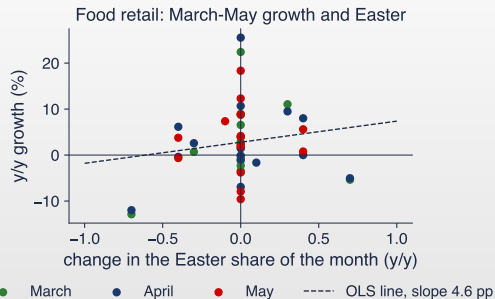
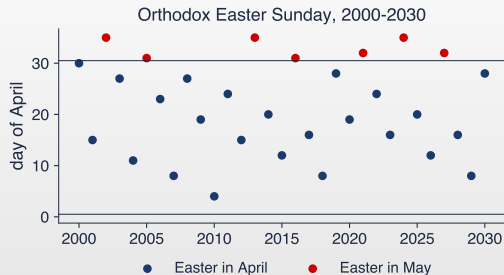
Painted Easter eggs from Romania

Photo: [jackmac34 \(Pixabay\)](#); CC0;
[Wikimedia Commons](#)

Regression with SARIMA errors

- $100 \ln y_t = \beta_E E_t + \beta_W WD_t + \text{outlier dummies} + u_t$, with $u_t \sim \text{SARIMA}(0, 1, 1)(0, 1, 1)_{12}$
 - ▶ E_t : the Easter regressor; WD_t : the number of working days in month t ; β_E, β_W : their effects, in %
 - ▶ the “regARIMA” model of X-13; estimated in one step by maximum likelihood (SARIMAX with exog)
 - ▶ ordinary least squares with SARIMA errors ignored would give wrong standard errors ([Chapter 3](#), spurious regression)
- Data: Eurostat retail trade volume, Romania, not adjusted, January 2010–July 2026: food (G47_FOOD) and all retail (G47)
 - ▶ dummies for April and May 2020 (lockdown)

Easter and Romanian retail trade



- Left: Orthodox Easter Sunday as a day of April (above 30: May); right: year-on-year growth of food retail in March-May against the change in the Easter share (2020–2021 left out)

Interpreting the Easter effect

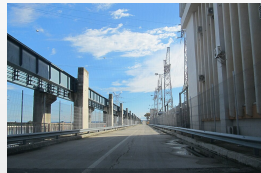
- Food retail: $\hat{\beta}_E = 5.9\%$ (se 2.3, $t = 2.6$): a month that holds the 10 days before Easter sells about 5.9% more food
 - ▶ one working day more: +0.59% (se 0.17); April 2020: -15%
- All retail: Easter 0.8% (se 1.9), not significant; working days +1.02% (se 0.15)
 - ▶ the Easter effect is in food, the working-day effect in everything
- The raw scatter is noisy (slope 4.6 percentage points): VAT changes and prices also move year-on-year growth; the regression separates the effects
- Easter fell in May in 2002, 2005, 2013, 2016, 2021, 2024: a forecaster who ignores this mistakes April and May

Recap: Fourier terms and calendar effects

- Fourier terms describe a seasonal pattern with $2K$ coefficients, for any period, even non-integer
- Working days and moving holidays are known in advance: model them as regressors
- Romanian food retail rises about 5.9% in the Easter month; total retail follows working days
- Regression with SARIMA errors: estimate regressors and dynamics together

Electricity load in Romania

- **Load:** the electricity consumed in the system, in MW (hourly averages); it must be matched by generation at every moment
 - ▶ forecasts are needed for every hour of the next day: day-ahead markets, reserves, grid planning
- **Data:** ENTSO-E (the European network of transmission system operators), “monthly hourly load values”, Romania
 - ▶ 39 406 hours, 2022–2026, Romanian winter time (UTC+2) to avoid the clock changes
- Load forecasting has its own competitions: GEFCom2014 ([Hong et al. \(2016\)](#))

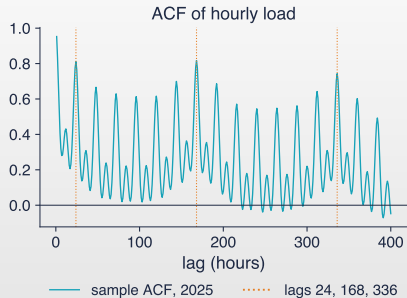
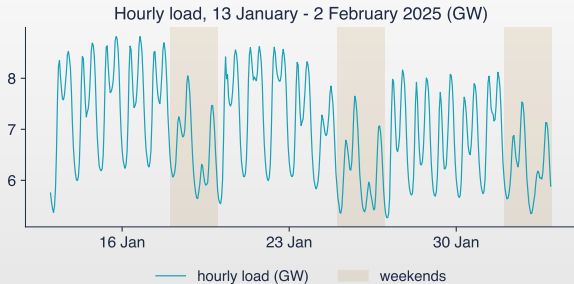


Iron Gates II hydropower plant on the Danube

Photo: Nenea hartia (2016); CC BY-SA 4.0;

[Wikimedia Commons](#)

Hourly load: the day and the week

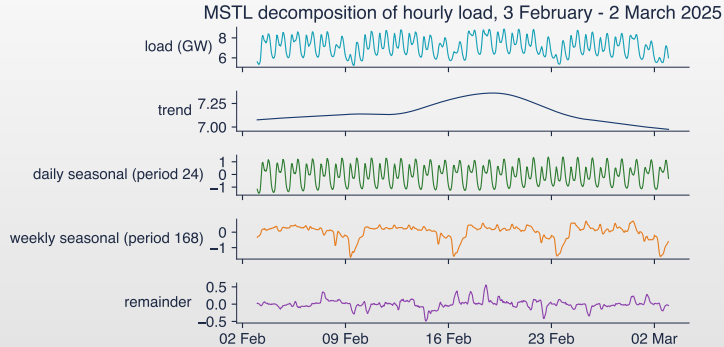


□ Left: three weeks of January 2025; right: sample ACF of all hours of 2025 up to lag 400

Interpreting the hourly load

- Daily cycle: lowest around 3:00 (5.3 GW on average in 2025), highest around 19:00 (7.3 GW)
 - ▶ ACF 0.81 at lag 24, 0.43 at lag 12
- Weekly cycle: weekends are lower; ACF 0.82 at lag 168 (one week), above the lag-24 value
 - ▶ the same hour one week ago is the best single predictor: Monday is like last Monday, not like Sunday
- And an annual cycle on top (heating and air conditioning): three periods, 24, 168 and about 8766 hours
- SARIMA is not enough: it has one integer period s , while here $s = 24, 168$ and 8766 act at once
 - ▶ a polynomial in L^{168} is slow and unstable, 365.25 is not an integer, and holidays break the weekly pattern
 - ▶ answers: MSTL, the dynamic harmonic regression (DHR), TBATS, Prophet

MSTL: one seasonal component per period

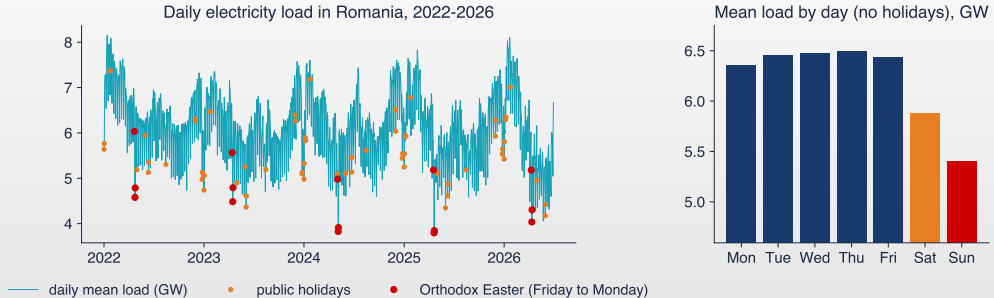


- MSTL (Bandara, Hyndman and Bergmeir (2025)) applies STL (Chapter 0) repeatedly, once for each period (24 and 168 hours), until the components settle

Interpreting the MSTL decomposition

- ▣ Of the variance around the trend: 70% daily component, 29% weekly component, 2% remainder
 - ▶ range of the daily component in the first week: 2.8 GW; of the weekly one: 2.1 GW
- ▣ The weekly component is flat Monday to Friday and drops on Saturday and more on Sunday
- ▣ The remainder is small but not white noise: cold spells and holidays; a forecasting model adds dynamics and regressors

Daily load and public holidays



- Daily mean load, 2022–2026, 1 642 days; Romanian public holidays marked; right: mean by day of the week, holidays excluded

Interpreting the daily load

- Week: Saturday 9% and Sunday 17% below Wednesday
 - ▶ Year: January 25% above May; a smaller summer peak (air conditioning)
- Holidays: the lowest day of the sample is Easter Sunday, 20 April 2025 (3.79 GW, mean 6.18 GW)
 - ▶ the four Easter days (Friday to Monday) average 4.62 GW; Christmas to New Year is also low
- The level falls over 2022–2026; possible causes: the price shock of 2022 and rooftop solar panels that cover part of the demand behind the meter

A dynamic harmonic regression for daily load

- $y_t = \sum_{k=1}^4 \left[\alpha_k \sin \frac{2\pi kt}{365.25} + \beta_k \cos \frac{2\pi kt}{365.25} \right] + \sum_j \delta_j D_{jt}^{\text{day}} + \lambda_1 H_t + \lambda_2 E_t + \lambda_3 X_t + u_t$
 - ▶ α_k, β_k : annual Fourier coefficients; δ_j : effect of weekday j against Sunday; $\lambda_1, \lambda_2, \lambda_3$: effects of the special days, in GW
 - ▶ annual Fourier terms; weekday dummies (Sunday is the base); H public holiday, E Easter (Friday to Monday), X 24 December – 2 January
 - ▶ $u_t \sim \text{ARIMA}(1,1,1)$, chosen by AIC: **dynamic harmonic regression** (DHR, [FPP3, Section 10.5](#))
- Estimates, data to the first forecast origin (30 June 2025), in GW:
 - ▶ Wednesday +1.08, Monday +0.88, Saturday +0.50 relative to Sunday
 - ▶ public holiday −0.38 (se 0.02), Easter −0.58 more, Christmas–New Year −0.61
 - ▶ errors: AR 0.80, MA −0.98, $\hat{\sigma} = 186$ MW; Ljung–Box(14) $p < 0.001$: some dynamics left

Recap: Multiple seasonality

- Hourly load has daily, weekly and annual cycles; daily load has weekly and annual cycles
- SARIMA handles one integer period; MSTL decomposes several
- Holidays, Easter most of all, are the largest deviations: they need regressors
- DHR = Fourier terms + calendar dummies + ARIMA errors: flexible and transparent

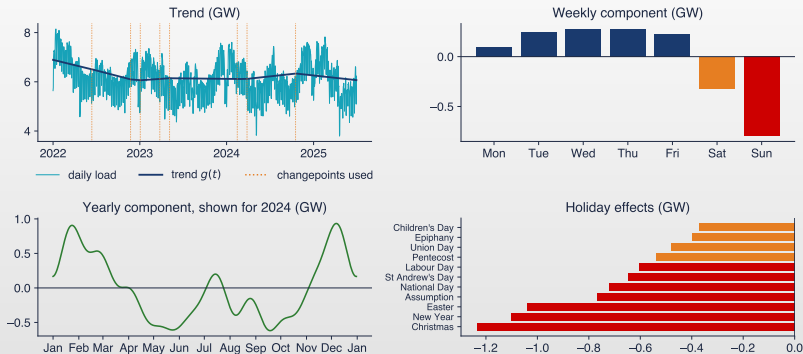
TBATS

- De Livera, Hyndman and Snyder (2011): exponential smoothing (ETS, [Chapter 0](#)) for complex seasonality; the name lists its parts
 - ▶ trigonometric seasonality (T), Box–Cox transformation (B), ARMA errors (A), trend (T), seasonal components (S)
- $y_t^{(\omega)} = \ell_{t-1} + \phi b_{t-1} + \sum_i s_{t-1}^{(i)} + d_t$, d_t ARMA(p, q); level and trend updated as in Holt's method
 - ▶ $y_t^{(\omega)}$: the series after a Box–Cox transformation with parameter ω ; ℓ_t : level; b_t : slope; ϕ : damping factor; $s_t^{(i)}$: seasonal component i
 - ▶ each period m_i has K_i harmonics $s_{j,t}^{(i)}$, each a rotating pair (cosine, sine) updated by the errors: a Fourier pattern that can evolve
- Automatic choices by AIC: Box–Cox or not, trend or damped trend or none, the K_i , the ARMA orders
 - ▶ daily load to June 2025, periods 7 and 365.25 (no Box–Cox, no trend): $K = 3$ weekly and $K = 9$ annual harmonics, ARMA(1,2), 28 states
- Limits: no regressors, hence no holidays or Easter; slow on long series; Python package `tbats`

Prophet

- Taylor and Letham (2018) (Meta): a regression in time, $y(t) = g(t) + s(t) + h(t) + \varepsilon_t$
 - ▶ $g(t)$: piecewise-linear (or logistic) trend whose slope may change at **changepoints**; a Laplace prior shrinks most changes to zero
 - ▶ $s(t)$: Fourier terms, by default $K = 10$ for the year and $K = 3$ for the week
 - ▶ $h(t)$: one effect per holiday (with optional windows around it), from a list of dates
- Estimated by Stan (maximum a posteriori); intervals simulate future changepoints and the noise
 - ▶ no autoregressive part: today's surprise does not move tomorrow's forecast
- Built for many business series with strong calendar effects and missing data; Python package prophet

Prophet components for daily load



□ Prophet with Romanian public holidays, daily load 2022 – June 2025

Interpreting the Prophet components

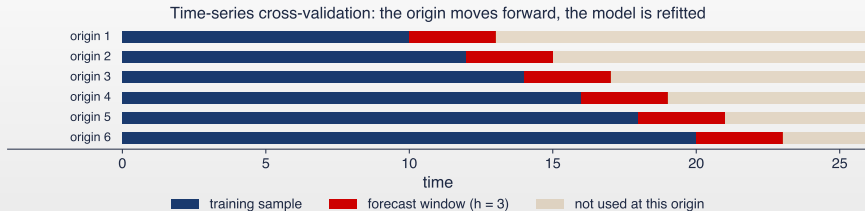
- Trend from 6.89 to 6.07 GW, flatter after 2023; 25 candidate changepoints, most shrunk to zero
 - ▶ weekly: Wednesday +0.27, Saturday -0.32, Sunday -0.80 GW
- Yearly: highest in early December (+0.93 GW, 7 December 2024), lowest in mid-September (-0.62 GW, 18 September 2024), shown for 2024
- Holidays: Christmas -1.24, New Year -1.10, Easter -1.04 GW; Children's Day only -0.37 GW
- Each component is readable and can be checked against what we know about the economy

Recap: DHR, TBATS and Prophet side by side

	DHR	TBATS	Prophet
Seasonality	Fourier, fixed	Fourier, evolving	Fourier, fixed
Several periods	yes	yes	yes
Holidays, regressors	yes	no	yes
Short-run dynamics	ARMA errors	ARMA errors	none
Trend	through the errors (ARIMA)	local (Holt)	piecewise linear
Speed	fast	slow	fast

- All three are regressions on sines and cosines at heart; they differ in what else they model
- Which forecasts best is an empirical question: the next section answers it out of sample

Time-series cross-validation



- ❑ Rolling forecast origin ([Tashman \(2000\)](#); [FPP3, Section 5.10](#)): fit on the data up to the origin, forecast h steps, move the origin, refit
- ❑ Expanding window (shown) or sliding window of fixed length; never use future data to fit

Interpreting the design of our experiment

- Daily load: 26 origins every 14 days, from 30 June 2025 to 15 June 2026; horizon $h = 14$ days
 - ▶ one year of out-of-sample forecasts, every season and every holiday, Easter 2026 included
- Seven methods, all refitted at each origin
 - ▶ seasonal naive (period 7); ETS (Holt–Winters, damped trend, period 7); SARIMA(2,0,1)(0, 1, 1)₇; DHR; TBATS; Prophet
 - ▶ Combination: the simple average of the five model forecasts (all except the seasonal naive)
- The orders of SARIMA and of the DHR errors are chosen once, by AIC, on the data before the first origin: no peeking at the test period
- Cross-validation that ignores time order (random folds) is valid only for pure autoregressions with white-noise errors (Bergmeir, Hyndman and Koo (2018))

The Diebold–Mariano test

- Diebold and Mariano (1995): is the difference in accuracy between two forecasts larger than chance?

- ▶ e_{1t} , e_{2t} : the errors of the two forecasts; L : the loss function
- ▶ loss differential $d_t = L(e_{1t}) - L(e_{2t})$, $L(e) = e^2$ or $|e|$; $H_0: E[d_t] = 0$

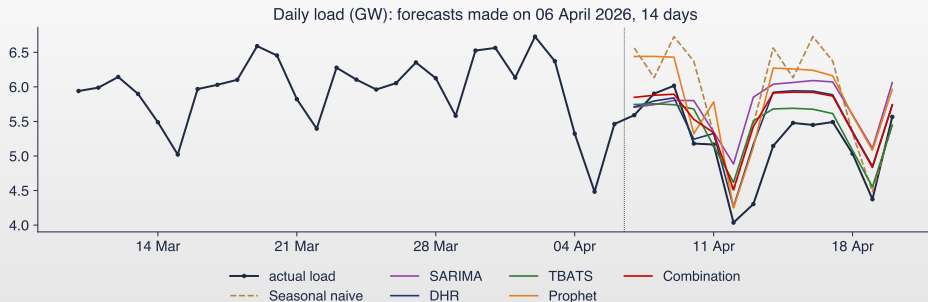
- $$DM = \frac{\bar{d}}{\sqrt{\hat{V}/n}}, \quad \hat{V} = \hat{\gamma}_d(0) + 2 \sum_{k=1}^{h-1} \hat{\gamma}_d(k)$$

- ▶ $\bar{d}$: the mean of the n differentials; $\hat{\gamma}_d(k)$: their autocovariance at lag k ; a negative DM favours forecast 1
- ▶ h -step errors overlap, so d_t is autocorrelated up to lag $h - 1$
- ▶ asymptotically $N(0, 1)$; small samples: $HLN = \sqrt{\frac{n+1-2h+h(h-1)/n}{n}}$ DM, compared with $t(n-1)$ (Harvey, Leybourne and Newbold (1997))
- Our design: one value per origin, d_j = average loss difference over the 14-day window; the windows do not overlap, so $h = 1$ in the formula
 - ▶ the test compares forecasts, not models: re-estimation noise is part of what is tested

Worked example: a Diebold–Mariano test

- Five windows; window-average differences in squared error, method 1 minus method 2:
 $-0.30, 0.10, -0.50, -0.20, -0.40$
 - ▶ $\bar{d} = -0.26$: method 1 has the smaller losses in four windows out of five
- $\hat{\gamma}_d(0) = \frac{1}{5} \sum (d_j - \bar{d})^2 = 0.042$
 - ▶ $DM = -0.26 / \sqrt{0.042/5} = -2.82$; $HLN = \sqrt{4/5} \cdot DM = -2.53$
- $|-2.53| < t_{0.975}(4) = 2.78$ ($p = 0.065$): equal accuracy is not rejected at 5%
 - ▶ the normal critical value 1.96 would have rejected: with few windows, the correction matters

Forecasts from one origin, across Easter 2026

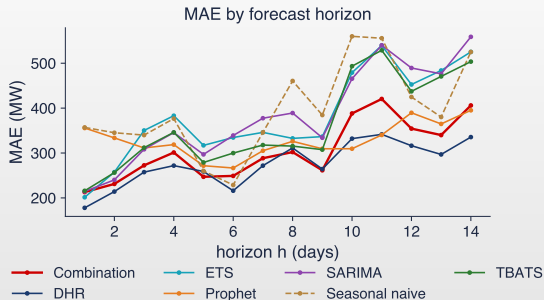
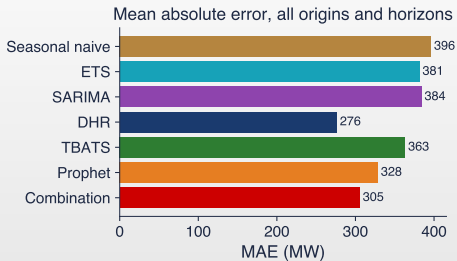


□ Origin 6 April 2026, 14 days ahead; Orthodox Easter Sunday was 12 April 2026

Interpreting one forecast window

- MAE in this window (MW): ETS 299, TBATS 309, DHR 344, Combination 394, SARIMA 584, Prophet 616, seasonal naive 686
 - ▶ the seasonal naive method copies the previous normal week into Easter week
- Every model sees the Sunday dip; DHR and Prophet also lower Good Friday and Easter Monday, which the weekly models treat as working days
- After Easter the load stays below every forecast for the rest of the week; ETS and TBATS, whose level follows the latest days, lose least here
- One window is anecdote; the next slides average over all 26 windows

Cross-validation results for daily load



26 origins $\times$ 14 horizons for each method; left: overall MAE; right: MAE by horizon

Interpreting the cross-validation results

Method	MAE	RMSE	MASE	DM against s. naive	DM against DHR
Seasonal naive	396	529	1.28	—	4.42 (< 0.001)
ETS	381	492	1.23	−0.80 (0.431)	3.08 (0.005)
SARIMA	384	477	1.24	−1.23 (0.231)	3.29 (0.003)
DHR	276	350	0.89	−4.42 (< 0.001)	—
TBATS	363	468	1.18	−1.50 (0.145)	3.02 (0.006)
Prophet	328	412	1.06	−2.99 (0.006)	2.76 (0.011)
Combination	305	380	0.99	−3.82 (< 0.001)	1.89 (0.070)

- MAE and RMSE in MW; MASE ([Hyndman and Koehler \(2006\)](#)) = MAE / 309 MW, the in-sample MAE of the weekly seasonal naive method; DM: HLN statistic (p-value) on window-average squared errors, row method minus column method
- DHR wins; with the combination, it is the only method with MASE below 1; Prophet is the second single model: the two that know the holidays
- ETS, SARIMA and TBATS are not significantly better than the seasonal naive method

A second test: Romanian GDP

Method	MAE, $h = 1$	MAE, $h = 4$	without 2019–2020, $h = 1$	without 2019–2020, $h = 4$
SARIMA (airline)	1.69	2.70	1.32	1.89
ETS	1.76	2.84	1.41	2.44
Seasonal naive	3.81	3.74	3.80	3.49
Combination (SARIMA, ETS)	1.70	2.65	1.34	2.02

- 100 ln Y, errors in %; 51 quarterly origins, 2012Q4–2025Q2, expanding window; 43 origins without 2019–2020
- DM (HLN), SARIMA against ETS: $h = 1$: -1.46 ($p = 0.150$); $h = 4$: 0.50 ($p = 0.620$)
- SARIMA against the seasonal naive method: $h = 1$: -4.00 ($p < 0.001$); $h = 4$: -0.54 ($p = 0.591$)
- One year ahead, the pandemic errors dominate and no difference is significant; without them, SARIMA has the smallest MAE at $h = 4$

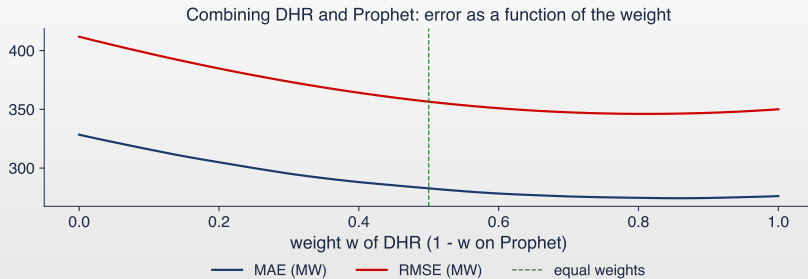
Recap: Forecast evaluation

- Evaluate out of sample with rolling origins; choose model orders before the test period
- MASE compares with the seasonal naive method and across series
- The DM test (with the HLN correction) says whether a gain is more than chance
- Daily load: the models that know the calendar (DHR, Prophet) win; for GDP, SARIMA and ETS are close

Why combining forecasts works

- **Bates and Granger (1969)**: a weighted average of two unbiased forecasts, $f_c = wf_1 + (1 - w)f_2$
 - ▶ w : the weight of forecast 1; σ_1, σ_2 : the standard deviations of the two errors; ρ : their correlation
 - ▶ $MSE(w) = w^2\sigma_1^2 + (1 - w)^2\sigma_2^2 + 2w(1 - w)\rho\sigma_1\sigma_2$
 - ▶ best weight $w^* = \frac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$; the gain is large when ρ is small
- **Example**: $\sigma_1 = 1, \sigma_2 = 1.2, \rho = 0.75$: $MSE(0.5) = 1.060$; $w^* = 0.84$, $MSE(w^*) = 0.984$
 - ▶ highly correlated errors: the equal-weight average is worse than the better forecast ($\sigma_1^2 = 1$), and even the best weight gains little
- **The forecast combination puzzle** (**Stock and Watson (2004)**; **Smith and Wallis (2009)**): estimated weights rarely beat equal weights
 - ▶ the weights must be estimated, and their sampling error eats the theoretical gain

Combining DHR and Prophet



□ Cross-validation errors of the daily load; w is the weight of DHR

Interpreting the combination

- MAE: Prophet alone ($w = 0$) 328 MW, DHR alone ($w = 1$) 276 MW, equal weights 283 MW
 - ▶ best in hindsight: $w = 0.86$, MAE 274 MW, a small gain chosen after seeing the test data
- The errors of the two methods are strongly correlated ($\rho = 0.75$): both miss the same cold spells
- The five-model average (MAE 305) is second only to DHR and better than four of its five members: cheap insurance when the winner is not known in advance
- In the M4 competition ([Makridakis, Spiliotis and Assimakopoulos \(2020\)](#)), most of the best methods were combinations; review: [Wang et al. \(2023\)](#)

Possible contribution of AI

- ▣ **Code:** a first draft of a rolling-origin loop that refits SARIMA, DHR, TBATS and Prophet and stores the errors by horizon
- ▣ **Calendars:** lists of holidays and moving feasts for other countries, to be checked against official sources
- ▣ **Explanation:** a second derivation of the ACF of the airline model, or of the eventual forecast function
- ▣ Example prompt
 - ▶ Write Python code that forecasts daily electricity load 14 days ahead with a regression on annual Fourier terms, weekday dummies and Romanian public holidays (Orthodox Easter from dateutil) with ARMA errors, and evaluates it against SARIMA by rolling origins and a Diebold-Mariano test.

Checks you must run

- ▣ The Easter date: Orthodox, not Western (EASTER_ORTHODOX); in 2024 the two differ by five weeks
- ▣ No look-ahead: features, model orders and holiday lists must be available at each origin
- ▣ The sign conventions of θ and Θ in the software, and the degrees of freedom of Ljung–Box
- ▣ Adjusted or unadjusted data: SARIMA on NSA; growth rates on the right version
- ▣ That the DM test uses losses from the same windows for both methods, and the HLN correction for small n
- ▣ Every cited reference: it must exist; check the DOI

Key takeaways

- ▣ Seasonality repeats with a known period; it can be deterministic (dummies, Fourier) or stochastic (Δ_s)
- ▣ SARIMA multiplies regular and seasonal polynomials; the airline model fits many series with two parameters
- ▣ Official data: SCA for short-run growth, NSA with calendar regressors for SARIMA forecasting
- ▣ Calendar effects (working days, Orthodox Easter, holidays) are known in advance and often matter more than the model class
- ▣ Multiple seasonality: MSTL, DHR, TBATS, Prophet; on Romanian daily load, DHR wins out of sample
- ▣ Judge forecasts by rolling origins, MASE and the DM test; combine them when the winner is uncertain

Key formulas

Quantity	Formula
Seasonal difference	$\Delta_s y_t = (1 - L^s)y_t, \quad \Delta \Delta_s = 1 - L - L^s + L^{s+1}$
SARIMA	$\phi(L)\Phi(L^s)(1 - L)^d(1 - L^s)^D y_t = \theta(L)\Theta(L^s)\varepsilon_t$
Airline	$\rho_1 = \frac{\theta}{1+\theta^2}, \quad \rho_s = \frac{\Theta}{1+\Theta^2}, \quad \rho_{s\pm 1} = \rho_1 \rho_s$
Fourier	$\sum_{k=1}^K [\alpha_k \sin(2\pi kt/m) + \beta_k \cos(2\pi kt/m)]$
Prophet	$y(t) = g(t) + s(t) + h(t) + \varepsilon_t$
MASE	$\frac{1}{h} \sum e_{T+j} / \frac{1}{T-m} \sum y_t - y_{t-m} $
DM	$\bar{d} / \sqrt{\hat{V}/n}, \quad d_t = L(e_{1t}) - L(e_{2t})$
Combination	$w^* = \frac{\sigma_2^2 - \rho \sigma_1 \sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho \sigma_1 \sigma_2}$

Self-assessment

- **Question:** how many parameters does $\text{SARIMA}(1, 1, 1)(0, 1, 1)_{12}$ have, and at which lags does ε appear?
 - ▶ **Answer:** $\phi, \theta, \Theta, \sigma^2$, four; ε at lags 0, 1, 12, 13 (the last with coefficient $\theta\Theta$)
- **Question:** unadjusted GDP falls by 40% from Q4 to Q1. Is this a recession?
 - ▶ **Answer:** no; it is the seasonal pattern; look at the SCA series or at year-on-year growth
- **Question:** a DM test with 26 windows gives $\text{HLN} = -1.93$. Is the first forecast significantly better at 5%?
 - ▶ **Answer:** no: $|-1.93| < t_{0.975}(25) = 2.06$; it is better on average, but not significantly
- **Project idea:** forecast Romanian hourly electricity load with a DHR (daily, weekly and annual Fourier terms, holidays, temperature) against TBATS, Prophet and a combination, by rolling origins and DM tests
 - ▶ Next: [Chapter 5](#), conditional volatility: ARCH and GARCH models

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